ADMITFLOW – FULL BACKEND SYSTEM DOCUMENTATION

🧠 1. SYSTEM OVERVIEW

AdmitFlow is an AI-powered admission automation platform that:

- Captures leads
- Qualifies them via chat/WhatsApp
- Uses AI voice agents for calls
- Converts leads into admissions

Core Idea:

AdmitFlow acts as a Prompt Compiler + Decision Engine, while Smallest.ai acts as the Execution Engine

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🔄 2. END-TO-END FLOW

👨‍💼 Admin Flow

1. Admin creates AI Agent
2. Adds:
    - Knowledge Base (courses, fees, FAQs)
    - Intents (user behavior)
    - Behavior Rules
    - Pitch Script
3. Clicks Generate Preview
4. Reviews final AI prompt
5. Clicks Deploy Agent

👤 User (Lead) Flow

1. Lead enters system (form / WhatsApp)
2. AI chats & qualifies lead
3. Behavior tracked (interest, hesitation, etc.)
4. If conditions met → AI Call triggered
5. Call happens via Smallest.ai

6. Result stored (transcript, outcome, score)

🧱 3. DATABASE SCHEMA

🏢 organizations

- id (UUID, PK)
- name
- created_at

🤖 ai_agents

- id (UUID, PK)
- organization_id (FK)
- name
- status (draft / active)
- smallest_agent_id
- tone
- language
- created_at

📚 knowledge_base

- id (UUID, PK)
- agent_id (FK)
- type (course / faq / general)
- title
- content (JSONB)
- created_at

🧠 intents

- id (UUID, PK)
- agent_id (FK)
- name
- triggers (JSONB)
- response
- goal
- tone

- created_at

⚙️ behavior_rules

- id (UUID, PK)
- agent_id (FK)
- condition_type
- condition_value
- action_type
- action_config (JSONB)
- priority
- created_at

🔁 call_flows

- id (UUID, PK)
- agent_id (FK)
- opening
- closing
- steps (JSONB)
- created_at

🧾 scripts

- id (UUID, PK)
- agent_id (FK)
- generated_script
- modified_script
- status (draft / approved)
- additional_instructions
- created_at

🧪 prompt_logs

- id (UUID, PK)
- agent_id (FK)
- final_prompt
- version
- created_at

👩‍💼 leads

- id (UUID, PK)
- organization_id (FK)
- name
- phone
- status
- score
- decision_state
- created_at

💬 conversations

- id (UUID, PK)
- lead_id (FK)
- message
- sender (user / ai)
- channel (whatsapp / chat)
- intent_detected
- created_at

🧠 behavior_events

- id (UUID, PK)
- lead_id (FK)
- event_type
- value
- score_impact
- created_at

📞 calls

- id (UUID, PK)
- lead_id (FK)
- agent_id (FK)
- call_status
- duration
- transcript
- created_at

⏳ followups

- id (UUID, PK)
- lead_id (FK)
- scheduled_at
- channel
- status
- created_at

🔗 4. RELATIONSHIPS

```
organizations

└── ai_agents



ai_agents

├── knowledge_base

├── intents

├── behavior_rules

├── call_flows

├── scripts

├── prompt_logs

└── calls



leads
```

```
├── conversations

├── behavior_events

├── calls

└── followups
```

⚙️ 5. BACKEND API STRUCTURE

🔐 Auth

- POST /auth/register
- POST /auth/login

🤖 Agents

- POST /agents
- GET /agents/:id
- PUT /agents/:id
- DELETE /agents/:id

📚 Knowledge Base

- POST /agents/:id/knowledge
- GET /agents/:id/knowledge
- PUT /knowledge/:id
- DELETE /knowledge/:id

🧠 Intents

- POST /agents/:id/intents
- GET /agents/:id/intents
- PUT /intents/:id
- DELETE /intents/:id

⚙️ Behavior Rules

- POST /agents/:id/rules
- GET /agents/:id/rules
- PUT /rules/:id
- DELETE /rules/:id

🧾 Script

- POST /agents/:id/script
- GET /agents/:id/script
- PUT /script/:id
- POST /script/:id/approve

🧠 Prompt System (CORE)

- GET /agents/:id/preview-prompt
- POST /agents/:id/deploy

👥 Leads

- POST /leads
- GET /leads
- PUT /leads/:id

💬 Conversations

- POST /conversations
- GET /leads/:id/conversations

📞 Calls

- POST /calls/trigger
- POST /calls/webhook

🧠 6. PROMPT BUILDER (CORE ENGINE)

```
async function buildPrompt(agentId) {

 const knowledge = await getKnowledge(agentId);
```

```
 const intents = await getIntents(agentId);

 const rules = await getRules(agentId);

 const script = await getApprovedScript(agentId);



 return `

You are a human-like admission counselor.



======================

KNOWLEDGE

======================

${knowledge}



======================

INTENTS

======================

${intents}



======================

CALL SCRIPT
```

```
======================

${script}



======================

RULES

======================

${rules}



======================

INSTRUCTIONS

======================

- Be natural

- Be concise

- Sound human

`;

}
```

🚀 7. DEPLOY FLOW

1. Admin clicks Deploy
2. Backend builds prompt
3. Send to Smallest.ai

4. Save `smallest_agent_id`
5. Agent becomes active

🔥 8. FINAL OUTPUT EXAMPLE

```
You are a human-like admission counselor.



KNOWLEDGE:

B.Tech AI, Fees: 2L



INTENTS:

If user says expensive → explain EMI



SCRIPT:

Hi, I saw you were interested...



RULES:

If user delays → schedule follow-up
```

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🧠 Intents

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